

Robust Beamforming Optimization for Cell-Free ISAC Systems Mathematical Derivation

1 Problem Formulation

Target-Centric Dedicated Sensing Associations. The binary variable $b_{mp} \in \{0, 1\}$ is a sensing-cluster admission indicator: it specifies whether AP m is authorized to allocate dedicated sensing resources to target p . Let $\mathbf{S}_p \succeq \mathbf{0}$ denote the covariance of the corresponding waveform. When $b_{mp} = 1$, AP m may allocate transmit power to $\mathbf{S}_p$; when $b_{mp} = 0$, its corresponding covariance block is forced to zero. Thus, $b_{mp} = 1$ is an authorization rather than a minimum-power guarantee, and realized participation is quantified from the nonzero sensing-power blocks after optimization. The indicator does not restrict communication beamforming, which remains globally cooperative. The per-AP transmit-hardware constraint is

$$\text{tr}(\mathbf{E}_m \mathbf{R}_X) \leq P_{\max}, \quad \forall m \in \mathcal{M}. \quad (1)$$

The association constraint and the dedicated sensing resource constraint are

$$\sum_{m=1}^M b_{mp} = N_{\text{req}}, \quad \text{tr}(\mathbf{E}_m \mathbf{S}_p) \leq P_{\max} b_{mp}, \quad \forall m, p. \quad (2)$$

The cardinality constraint specifies a fixed sensing-cluster size for each target. If N_{req} is intended as an upper bound rather than a fixed budget, the equality should be replaced by $\leq N_{\text{req}}$ and accompanied by an explicit fronthaul or processing cost.

The total transmit covariance is $\mathbf{R}_X = \sum_{k=1}^K \mathbf{W}_k + \sum_{p=1}^P \mathbf{S}_p$. To retain a known and target-specific probing signal at the sensing receiver, the PCRB of target p is evaluated only from $\mathbf{S}_p$:

$$\mathbf{J}_p^{\text{data}}(\mathbf{S}_p) = \frac{2}{\sigma_s^2} \text{Re}\{\mathbf{D}_p^H \mathbf{S}_p \mathbf{D}_p\}. \quad (3)$$

Thus, the association variable controls the dedicated sensing resources that can contribute to the PCRB, whereas communication covariances do not receive PCRB credit. The constraints above are affine in the continuous covariance variables for fixed b_{mp} and linear in b_{mp} for fixed covariances. Unless receiver-side cancellation is explicitly assumed, the aggregate dedicated sensing covariance $\sum_p \mathbf{S}_p$ is included as interference in the communication SINR. The Sum Big-M power-gating formulation and derivations based on an FIM that is affine solely in $\mathbf{R}_X$ correspond to a different model and should be replaced by the present formulation.

The decision variables of the original problem (P1) are the per-AP beamforming vectors $\mathbf{w}_{m,k} \in \mathbb{C}^{N_t}$, the target-specific sensing covariance matrices $\mathbf{S}_p \in \mathbb{H}_+^{MN_t}$, and the AP-target soft-assignment weights $b_{mp} \in [0, 1]$, which are a continuous relaxation of the association indicator described above:

$$\min_{\substack{\{\mathbf{w}_{m,k}\}_{m \in \mathcal{M}, k \in \mathcal{K}}, \{\mathbf{S}_p\}_{p \in \mathcal{P}}, \\ \{b_{mp}\}_{m \in \mathcal{M}, p \in \mathcal{P}}}} \sum_{m=1}^M \sum_{k=1}^K \|\mathbf{w}_{m,k}\|^2 + \sum_{p=1}^P \text{tr}(\mathbf{S}_p) \quad (\text{P1-objective})$$

$$\text{s.t.} \quad \min_{\|\Delta \mathbf{h}_k\| \leq \epsilon_h, \|\hat{\mathbf{h}}_k\|} \frac{|(\hat{\mathbf{h}}_k + \Delta \mathbf{h}_k)^H \mathbf{w}_k|^2}{\sum_{j \neq k} |(\hat{\mathbf{h}}_k + \Delta \mathbf{h}_k)^H \mathbf{w}_j|^2 + (\hat{\mathbf{h}}_k + \Delta \mathbf{h}_k)^H (\sum_p \mathbf{S}_p) (\hat{\mathbf{h}}_k + \Delta \mathbf{h}_k) + \sigma_c^2} \geq \gamma_S^{\text{PoD}} \quad (\text{P1-C1})$$

$$\text{SINR}_{S,p} := \frac{\mathbf{g}_p^H \mathbf{S}_p \mathbf{g}_p}{\sigma_s^2} \geq \gamma_S^{\text{PoD}}, \quad \forall p \in \mathcal{P}, \quad (\text{P1-C2})$$

$$\text{tr}((\mathbf{J}_p^{\text{data}}(\mathbf{S}_p))^{-1}) \leq \Gamma_{\text{Track},p}, \quad \forall p \in \mathcal{P}, \quad (\text{P1-C3})$$

$$\text{tr}(\mathbf{E}_m \mathbf{S}_p) \leq P_{\max} b_{mp}, \quad \forall m \in \mathcal{M}, p \in \mathcal{P}, \quad (\text{P1-C4'a})$$

$$\text{tr}(\mathbf{E}_m \mathbf{R}_X) \leq P_{\max}, \quad \forall m \in \mathcal{M}, \quad (\text{P1-C4'b})$$

$$\sum_{m=1}^M b_{mp} = N_{\text{req}}, \quad \forall p \in \mathcal{P}^{\text{active}}, \quad (\text{P1-C5'})$$

$$\mathbf{S}_p \succeq \mathbf{0}, \quad \forall p \in \mathcal{P}, \quad (\text{P1-C6})$$

$$0 \leq b_{mp} \leq 1, \quad \forall m \in \mathcal{M}, p \in \mathcal{P}. \quad (\text{P1-C7'})$$

where $\mathbf{w}_k \triangleq [\mathbf{w}_{1,k}^T, \mathbf{w}_{2,k}^T, \dots, \mathbf{w}_{M,k}^T]^T \in \mathbb{C}^{MN_t}$ is the cooperative stacked beamforming vector from all APs to user k , obtained by stacking the per-AP variables; $\mathbf{R}_X \triangleq \sum_{k=1}^K \mathbf{w}_k \mathbf{w}_k^H + \sum_{p=1}^P \mathbf{S}_p$ (precisely defined in the lifted form of §2-A); and $\mathbf{E}_m \in \mathbb{R}^{MN_t \times MN_t}$ is the diagonal selection matrix that extracts the antenna block of AP m from $\mathbf{R}_X$. The FIM $\mathbf{J}_p^{\text{data}}$ in (P1-C3) takes the following original form for a multi-dimensional target state $\boldsymbol{\theta}_p \in \mathbb{R}^{N_\theta}$ (typical ISAC scenarios involve 2D/3D position, or angle+range with $N_\theta \in \{2, 3, 4\}$):

$$\mathbf{J}_p^{\text{data}}(\mathbf{S}_p) = \frac{2}{\sigma_s^2} \text{Re}\{\mathbf{D}_p^H \mathbf{S}_p \mathbf{D}_p\} \in \mathbb{H}_+^{N_\theta \times N_\theta}, \quad (4)$$

where $\mathbf{D}_p \triangleq \partial \mathbf{g}_p / \partial \boldsymbol{\theta}_p \in \mathbb{C}^{MN_t \times N_\theta}$ is the effective target-response derivative matrix. The equivalent MISO observation model uses the dedicated sensing covariance directly; hence (4) restates (3) and is affine in $\mathbf{S}_p$ only—the communication covariances $\{\mathbf{W}_k\}$ receive no PCRB credit. The CRB matrix is $\mathbf{C}_p = (\mathbf{J}_p^{\text{data}})^{-1}$ and (P1-C3) requires $\text{tr}(\mathbf{C}_p) \leq \Gamma_{\text{Track},p}$. This matrix trace-of-inverse constraint must be retained for $N_\theta \geq 2$; for $N_\theta = 1$, it reduces to the scalar condition $\mathbf{J}_p^{\text{data}} \geq 1/\Gamma$. The association gate (P1-C4'a) and the hardware ceiling (P1-C4'b) are both linear: (P1-C4'a) authorizes AP m to allocate dedicated sensing power to target p up to $P_{\max}$ when $b_{mp} = 1$ and forces the corresponding block of $\mathbf{S}_p$ to zero when $b_{mp} = 0$, without gating communication transmission, while (P1-C4'b) caps the total per-AP transmit covariance $\mathbf{R}_X$.

Problem (P1) is non-convex. Its non-convexity stems from four independent sources distributed across different constraints. The communication worst-case SINR constraint (P1-C1) carries a fractional quadratic structure: the numerator $|\mathbf{h}_k^H \mathbf{w}_k|^2$ and the denominator $\sum_{j \neq k} |\mathbf{h}_k^H \mathbf{w}_j|^2 + \mathbf{h}_k^H (\sum_p \mathbf{S}_p) \mathbf{h}_k + \sigma_c^2$ are both quadratic functions of the beamforming vectors and the sensing covariances, so their ratio is a non-convex function on the manifold where the denominator is positive [1]. Compounding this, the worst-case quantifier

$\forall \|\Delta \mathbf{h}_k\| \leq \epsilon_h \|\hat{\mathbf{h}}_k\|$ is semi-infinite, and the resulting set $\{\mathbf{w} : \min_{\Delta \mathbf{h} \in \mathcal{B}_\epsilon} \text{SINR}_k(\Delta \mathbf{h}) \geq \gamma_k\}$ is not closed under convex combinations. The position PCRB constraint (P1-C3) requires $\text{tr}((\mathbf{J}_p^{\text{data}})^{-1}) \leq \Gamma_{\text{Track},p}$; although the composite function $\text{tr}(\mathbf{J}^{-1})$ is strictly convex on $\mathbf{J} \succ \mathbf{0}$, so that (P1-C3) itself defines a convex set, the constraint involves the non-linear matrix inverse and must be transformed into an LMI via the Schur complement before standard SDP solvers can handle it. The binarity of the AP–target soft-assignment weights b_{mp} in (P1-C7’), equivalently rewritten as $g(b) = b - b^2 = 0$, manifests itself as a concave penalty since $g(b) \geq 0$ for $b \in [0, 1]$ and g is strictly concave. The association gate (P1-C4’a) and the per-AP hardware ceiling (P1-C4’b) are both linear: the left-hand sides are the linear trace terms $\text{tr}(\mathbf{E}_m \mathbf{S}_p)$ and $\text{tr}(\mathbf{E}_m \mathbf{R}_X)$ extracted by the diagonal selection matrix $\mathbf{E}_m$, and the right-hand sides are $P_{\max} b_{mp}$ (association-coupled) and $P_{\max}$ (standard hardware upper bound) respectively; both constraints are convex in the joint variable space $(\{\mathbf{w}\}, \{\mathbf{S}_p\}, b)$ (the box $0 \leq b_{mp} \leq 1$ is convex). The PSD cone $\mathbf{S}_p \succeq \mathbf{0}$ itself ((P1-C6)), the linear sensing SINR $\text{tr}(\mathbf{g}_p \mathbf{g}_p^H \mathbf{S}_p) \geq \gamma_s^{\text{PoD}} \sigma_s^2$ ((P1-C2)), and the box constraint $0 \leq b_{mp} \leq 1$ ((P1-C7’)) are individually convex, but because of the four non-convex/nonlinear sources above the overall problem remains a non-convex MINLP. In §2 we address each source by a dedicated step of the convexification chain (the rank-1 constraint that arises after covariance lifting is treated as a fifth source and is recovered jointly with the binarity of b_{mp} by the dual DC penalty SCA framework of §3; the binarity of b_{mp} is already in the (P1) domain as a box + DC penalty, hence is handled in §3 in the same SCA loop).

2 Solution of the Optimization Problem

This section solves the problem along the five non-convex/nonlinear sources identified at the end of §1. The SINR fraction and the semi-infinite uncertainty are converted into LMIs via cross-multiplication (§2-C step 1) followed by the S-Procedure (§2-C step 2); the trace of the PCRB matrix inverse is losslessly converted into an LMI by the Schur complement (§2-D); the binarity of the AP–target soft-assignment weights b_{mp} is handled by continuous relaxation plus a DC penalty added to the objective (§2-E), which shares a unified dual-DC-penalty SCA framework with the rank-1 constraint in §3.

A. Problem Transformation (P2)

To prepare for the semidefinite relaxation, we lift the vector variables in (P1) into covariance matrices, i.e. we define $\mathbf{W}_k = \mathbf{w}_k \mathbf{w}_k^H \in \mathbb{H}_+^{MN_t}$ and $\mathbf{R}_X \triangleq \sum_{k=1}^K \mathbf{W}_k + \sum_{p=1}^P \mathbf{S}_p$, where the target-specific sensing covariances $\mathbf{S}_p \in \mathbb{H}_+^{MN_t}$ were introduced in §1, yielding

(P2):

$$\min_{\{\mathbf{W}_k\}_{k \in \mathcal{K}}, \{\mathbf{S}_p\}_{p \in \mathcal{P}}, \{b_{mp}\}_{m \in \mathcal{M}, p \in \mathcal{P}}} \sum_{k=1}^K \text{tr}(\mathbf{W}_k) + \sum_{p=1}^P \text{tr}(\mathbf{S}_p) \quad (\text{P2})$$

$$\text{s.t.} \quad (\forall \|\Delta \mathbf{h}_k\| \leq \epsilon_h \|\hat{\mathbf{h}}_k\| : \text{tr}(\hat{\mathbf{H}}_k(\Delta \mathbf{h}_k) \mathbf{W}_k) - \gamma_k \sum_{j \neq k} \text{tr}(\hat{\mathbf{H}}_k(\Delta \mathbf{h}_k) \mathbf{W}_j) - \gamma_k \text{tr}(\hat{\mathbf{H}}_k(\Delta \mathbf{h}_k) \sum_p \mathbf{S}_p) \leq 0) \quad (\text{P2-C1})$$

$$\text{tr}(\mathbf{g}_p \mathbf{g}_p^H \mathbf{S}_p) \geq \gamma_S^{\text{PoD}} \sigma_s^2, \quad \forall p \in \mathcal{P}, \quad (\text{P2-C2})$$

$$\text{tr}((\mathbf{J}_p^{\text{data}}(\mathbf{S}_p))^{-1}) \leq \Gamma_{\text{Track}, p}, \quad \forall p \in \mathcal{P}, \quad (\text{P2-C3})$$

$$\text{tr}(\mathbf{E}_m \mathbf{S}_p) \leq P_{\max} b_{mp}, \quad \forall m \in \mathcal{M}, p \in \mathcal{P}, \quad (\text{P2-C4'a})$$

$$\text{tr}(\mathbf{E}_m \mathbf{R}_X) \leq P_{\max}, \quad \forall m \in \mathcal{M}, \quad (\text{P2-C4'b})$$

$$\sum_{m=1}^M b_{mp} = N_{\text{req}}, \quad \forall p \in \mathcal{P}^{\text{active}}, \quad (\text{P2-C5'})$$

$$\mathbf{W}_k \succeq \mathbf{0}, \quad \forall k \in \mathcal{K}, \quad (\text{P2-C6})$$

$$\mathbf{S}_p \succeq \mathbf{0}, \quad \forall p \in \mathcal{P}, \quad (\text{P2-C7})$$

$$\text{rank}(\mathbf{W}_k) = 1, \quad \forall k \in \mathcal{K}, \quad (\text{P2-C8})$$

$$0 \leq b_{mp} \leq 1, \quad \forall m \in \mathcal{M}, p \in \mathcal{P}. \quad (\text{P2-C9})$$

Here $\mathbf{J}_p^{\text{data}}(\mathbf{S}_p) = \frac{2}{\sigma_s^2} \text{Re}\{\mathbf{D}_p^H \mathbf{S}_p \mathbf{D}_p\} \in \mathbb{H}_+^{N_\theta \times N_\theta}$ is the data-dependent Fisher Information Matrix under the equivalent MISO sensing observation model, where $\mathbf{D}_p = \partial \mathbf{g}_p / \partial \boldsymbol{\theta}_p \in \mathbb{C}^{MN_t \times N_\theta}$ is the effective target-response derivative; $\hat{\mathbf{H}}_k(\Delta \mathbf{h}_k) \triangleq (\hat{\mathbf{h}}_k + \Delta \mathbf{h}_k)(\hat{\mathbf{h}}_k + \Delta \mathbf{h}_k)^H$. This model uses the dedicated sensing covariance $\mathbf{S}_p$ directly and does not posit a vectorized multi-receiver architecture. Note that each $\mathbf{S}_p$ in (P2) is a joint stacked covariance, so the dedicated waveform for target p may still be synthesized cooperatively across APs; the association gate (P2-C4'a) only authorizes which APs may allocate power to $\mathbf{S}_p$, while (P2-C2) sensing SINR and (P2-C3) PCRB are both evaluated from $\mathbf{S}_p$ alone.

(P1) and (P2) are strictly equivalent: the variable mapping $\mathbf{w}_k \leftrightarrow \mathbf{W}_k = \mathbf{w}_k \mathbf{w}_k^H$ is a bijection (the rank-1 constraint guarantees recoverability), and every constraint is rewritten equivalently through identities such as $\text{tr}(\mathbf{x} \mathbf{x}^H \mathbf{W}) = \mathbf{x}^H \mathbf{W} \mathbf{x}$. The box constraint (P2-C9) replaces the binary constraint (P1-C7'); introducing the DC penalty (detailed in §2-E) does not change the feasible region of (P2). The overall equivalence transformation introduces no relaxation gap.

Problem (P2) is a rewrite of (P1) in the lifted covariance domain. The covariance lifting preserves four of the six sources identified in §1: the SINR fractional structure in (P2-C1) (NC1); the soft-assignment equality and box constraints in (P2-C5')/(P2-C9) (NC3, softened); the semi-infinite worst-case quantifier (NC2), rewritten in the (P2) domain as a quadratic form in $\Delta \mathbf{h}_k$, to be handled by the S-Procedure in §2 Step 3; and the PCRB matrix inverse (NC6) in (P2-C3), which remains in the form $\text{tr}((\mathbf{J}_p^{\text{data}})^{-1}) \leq \Gamma_{\text{Track}, p}$, to be processed by the Schur complement in §2-D. The former NC5 beam-sensing bilinear coupling vanishes under the dedicated-covariance formulation, since (P2-C4'a) and (P2-C4'b) are both linear. Lifting introduces one new source of non-convexity: the rank-1 manifold constraint (NC4, (P2-C8)) requiring $\mathbf{W}_k = \mathbf{w}_k \mathbf{w}_k^H$. The PSD cone constraints (P2-C6) and (P2-C7) are themselves convex, and (P2-C2) (sensing SINR) becomes an affine function of $\mathbf{S}_p$ in the lifted domain through the identity $\mathbf{g}_p^H \mathbf{S}_p \mathbf{g}_p = \text{tr}(\mathbf{g}_p \mathbf{g}_p^H \mathbf{S}_p)$.

The residual non-convex/nonlinear sources in the (P2) domain are handled jointly by the convexification chain of §2 and the dual DC penalty SCA framework of §3.

B. Semi-Definite Relaxation

The NC4 in (P2) is addressed by SDR: the rank-1 manifold $\mathcal{F}_{\text{rank-1}}$ is itself non-convex, and SDR relaxes (P2-C8) to $\mathbf{W}_k \succeq \mathbf{0}$ only, i.e.

$$\mathbf{W}_k \succeq \mathbf{0}, \quad \forall k, \quad (\text{S1.1})$$

The feasible region is enlarged from $\mathcal{F}_{\text{rank-1}}$ to the positive semidefinite cone $\mathcal{F}_{\text{SDR}}$, and the SDR solution provides a lower bound on the original problem, $P_{\text{SDR}}^* \leq P_{\text{P2}}^*$. For the joint structure of per-user SINR + cell-free cooperation + robust uncertainty, neither the multi-group multicasting tightness condition $G_k \leq 2$ [2] nor the robust MISO condition $N_t \leq 2$ [3] directly applies, and Gaussian randomization must be used to recover a feasible rank-1 solution when $\text{rank}(\mathbf{W}_k^*) > 1$. Specifically, $L \in [100, 1000]$ Gaussian random vectors are generated and the candidate that satisfies the constraints and is optimal in the objective is selected; however, this post-processing has no tight performance-loss upper bound.

C. S-Procedure

This subsection handles the residual semi-infinite robust uncertainty in NC1 in two steps: first the original SINR robust inequality is cross-multiplied into a quadratic form (an equivalent rewrite of (P2-C1), no longer a fraction), and then the S-Procedure reduces the universal quantifier over $\Delta \mathbf{h}_k$ to an LMI in the S-Procedure relaxation variable μ_k .

Step 1 – Cross-multiplication. Fix any worst-case $\Delta \mathbf{h}_k$ satisfying $\|\Delta \mathbf{h}_k\| \leq \epsilon_h \|\hat{\mathbf{h}}_k\|$. Because $\sigma_c^2 > 0$ guarantees that the denominator is strictly positive, the SINR inequality can be cross-multiplied directly into the quadratic form

$$\text{tr}(\hat{\mathbf{H}}_k \mathbf{W}_k) - \gamma_k \sum_{j \neq k} \text{tr}(\hat{\mathbf{H}}_k \mathbf{W}_j) - \gamma_k \text{tr}\left(\hat{\mathbf{H}}_k \sum_p \mathbf{S}_p\right) \geq \gamma_k \sigma_c^2, \quad (\text{S2.1})$$

which is a strict equivalence rather than an approximation; strict positivity of the denominator guarantees that the implication holds in both directions. Since σ_c^2 contains no decision variable, the fraction can be eliminated without a first-order Taylor approximation, thus avoiding the accumulation of SCA iteration error.

Step 2 – S-Procedure. Restoring $\Delta \mathbf{h}_k$ as a variable and defining $\mathbf{A}_k \triangleq \frac{1}{\gamma_k} \mathbf{W}_k - \sum_{j \neq k} \mathbf{W}_j - \sum_p \mathbf{S}_p$, (P2-C1) is rewritten as a quadratic inequality in $\Delta \mathbf{h}_k$:

$$\tilde{f}_1(\Delta \mathbf{h}) = (\hat{\mathbf{h}} + \Delta \mathbf{h})^H \mathbf{A}_k (\hat{\mathbf{h}} + \Delta \mathbf{h}) - \sigma_c^2 \geq 0, \quad \forall \|\Delta \mathbf{h}_k\| \leq \epsilon_h \|\hat{\mathbf{h}}_k\|,$$

and the uncertainty set is described by $\tilde{f}_2(\Delta \mathbf{h}) = \epsilon_h^2 \|\hat{\mathbf{h}}_k\|^2 - \|\Delta \mathbf{h}_k\|^2 \geq 0$. For this single quadratic constraint on a norm ball, the universal quantifier admits, via the S-Procedure [1], a finite LMI equivalent condition:

Lemma 1. (*S-Procedure*). Let $\tilde{f}_m(\Delta \mathbf{h}) = (\Delta \mathbf{h})^H \mathbf{A}_m (\Delta \mathbf{h}) + 2\text{Re}\{(\Delta \mathbf{h})^H \mathbf{b}_m\} + c_m$, $m \in \{1, 2\}$. Then the implication “ $\tilde{f}_1(\Delta \mathbf{h}) \geq 0 \Rightarrow \tilde{f}_2(\Delta \mathbf{h}) \geq 0$ ” holds if and only if there exists $\mu \geq 0$ such that

$$\begin{bmatrix} \mathbf{A}_2 + \mu \mathbf{A}_1 & \mathbf{b}_2 + \mu \mathbf{b}_1 \\ \mathbf{b}_2^H + \mu \mathbf{b}_1^H & c_2 + \mu c_1 \end{bmatrix} \succeq \mathbf{0},$$

This is the original form of the S-Procedure [1, §4.3], which requires no additional hypotheses in the norm-ball setting where $\tilde{f}_2$ describes the convex set $\{\Delta \mathbf{h} : \|\Delta \mathbf{h}\|_2 \leq \epsilon_h \|\hat{\mathbf{h}}\|_2\}$.

Applying Lemma 1 directly to the implication $\tilde{f}_1 \geq 0 \Rightarrow \tilde{f}_2 \geq 0$ (where $\tilde{f}_2$ describes the norm-ball uncertainty set), the worst-case robust form of (P2-C1) can be written as the following LMI:

$$\exists \mu_k \geq 0 : \begin{bmatrix} \mathbf{A}_k + \mu_k \mathbf{I} & \mathbf{A}_k \hat{\mathbf{h}}_k \\ \hat{\mathbf{h}}_k^H \mathbf{A}_k & \hat{\mathbf{h}}_k^H \mathbf{A}_k \hat{\mathbf{h}}_k - \sigma_c^2 - \mu_k \epsilon_h^2 \|\hat{\mathbf{h}}_k\|^2 \end{bmatrix} \succeq \mathbf{0}. \quad (\text{S3.1})$$

The newly introduced $\mu_k \geq 0$ is the S-Procedure relaxation variable.

D. PCRB Constraint: Schur Complement LMI Transformation

The PCRB constraint (P1-C3) remains in the lifted domain as

$$\text{tr}((\mathbf{J}_p^{\text{data}})^{-1}) \leq \Gamma_{\text{Track},p}, \quad \forall p \in \mathcal{P}, \quad (\text{P2-C3})$$

where $\mathbf{J}_p^{\text{data}} \succ \mathbf{0}$ is an affine function of $\mathbf{S}_p$. The matrix-inverse map $\mathbf{J} \mapsto \mathbf{J}^{-1}$ is strictly matrix-convex on the positive-definite cone, and the trace operator $\text{tr}(\cdot)$ is a linear order-preserving map; therefore the composite $\text{tr}(\mathbf{J}^{-1})$ is a strictly convex function and the constraint (P2-C3) itself defines a convex set. However, the constraint involves the non-linear matrix inverse; to make it directly processable by an interior-point SDP solver, this subsection losslessly converts it into an LMI via the Schur complement.

Introduce an auxiliary Hermitian matrix variable $\mathbf{M}_p \in \mathbb{H}^{N_\theta \times N_\theta}$ (where N_θ is the dimension of the target parameter to be estimated). The original constraint can be split equivalently as

$$\mathbf{M}_p - (\mathbf{J}_p^{\text{data}})^{-1} \succeq \mathbf{0}, \quad \forall p \in \mathcal{P}, \quad (5a)$$

$$\text{tr}(\mathbf{M}_p) \leq \Gamma_{\text{Track},p}, \quad \forall p \in \mathcal{P}. \quad (5b)$$

Since $\mathbf{J}_p^{\text{data}} \succ \mathbf{0}$, by the Schur complement lemma, (5a) is equivalent to

$$\begin{bmatrix} \mathbf{M}_p & \mathbf{I} \\ \mathbf{I} & \mathbf{J}_p^{\text{data}} \end{bmatrix} \succeq \mathbf{0}, \quad \forall p \in \mathcal{P}. \quad (6)$$

Therefore, the PCRB constraint (P2-C3) is converted completely and without approximation into the linear matrix inequalities in $(\{\mathbf{S}_p\}, \{\mathbf{M}_p\})$:

$$\begin{bmatrix} \mathbf{M}_p & \mathbf{I} \\ \mathbf{I} & \mathbf{J}_p^{\text{data}} \end{bmatrix} \succeq \mathbf{0}, \quad \forall p \in \mathcal{P}, \quad (7a)$$

$$\text{tr}(\mathbf{M}_p) \leq \Gamma_{\text{Track},p}, \quad \forall p \in \mathcal{P}. \quad (7b)$$

This transformation introduces no relaxation gap and avoids outer SCA iterations.

E. AP–Target Soft Assignment: Continuous Relaxation with DC Penalty

This subsection completely replaces the path-loss heuristic AP selection of the original literature. The original formulation peels off the binary constraint $\{0, 1\}$ of (P2-C8),

sorts APs by the large-scale fading $PL(d_{m,p})$, fixes the top- N_{req} APs as the serving set $\mathcal{M}_p$, and solves the inner SDP over the fixed set; this procedure is a K -medoid variant and is NP-hard, the outer heuristic loop cannot provide any global optimality guarantee, and is inconsistent with the rigorous DC-penalty SCA framework used for the rank-1 constraint in §3. The rigorous replacement given here is as follows: merge the box constraint $0 \leq b_{mp} \leq 1$ of (P2-C9) with the binarity $b_{mp} \in \{0, 1\}$ into the equivalent form

$$0 \leq b_{mp} \leq 1, \quad (\text{E.1a})$$

$$g(b_{mp}) \triangleq b_{mp} - b_{mp}^2 = 0. \quad (\text{E.1b})$$

For $b \in [0, 1]$, $g(b) = b - b^2 \geq 0$ holds and g is a concave function of b (since $g''(b) = -2 < 0$); reformulating the equality constraint (E.1b) as a penalty term added to the objective gives

$$\min \underbrace{\sum_{k=1}^K \text{tr}(\mathbf{W}_k) + \sum_{p=1}^P \text{tr}(\mathbf{S}_p)}_{\text{original objective}} + \eta_b \underbrace{\sum_{m \in \mathcal{M}, p \in \mathcal{P}} g(b_{mp})}_{=\sum_{m,p} (b_{mp} - b_{mp}^2)} \quad \text{s.t. (P2-C1) } \sim \text{(P2-C9)} \setminus \{(\text{P2-C9})\}, \quad (\text{P2-DCP})$$

where $\eta_b \geq 0$ is the penalty factor; the objective exhibits a DC structure (“convex function + concave function $\times \eta_b$ ”) and the box constraints (P2-C9) are kept in their convex form, but the term $-\eta_b b_{mp}^2$ in the objective is concave in b_{mp} , so (P2-DCP) is not a convex optimization problem and this non-convexity is handled jointly with the rank-1 constraint by the dual DC penalty SCA framework in §3. For $b_{mp} \in \{0, 1\}$, $g(b_{mp}) = 0$; for $b_{mp} \in (0, 1)$, $g(b_{mp}) > 0$ and the penalty grows with η_b . A finite penalty can reduce but does not by itself prove that this residual vanishes; exact recovery must be established by the residual test and fixed-assignment recovery in §3. After introducing the η_b concave penalty, (P2-DCP) has exactly the same mathematical structure as the rank-1 penalty $\rho(\mathbf{W}_k) = \text{tr}(\mathbf{W}_k) - \lambda_{\max}(\mathbf{W}_k)$ introduced in §3-A—both are “convex – convex” DC forms—so the SCA framework designed in §3 can carry both penalties simultaneously. A large-scale-fading heuristic may be used as a warm start, but for this non-convex SCA it can affect both convergence speed and the stationary point reached.

F. Consolidation (P3): Final Form with Dual DC Penalties

Applying the SDR step of §-B ((S1.1)), the S-Procedure step of §-C ((S2.1) + (S3.1)), the Schur complement of §-D ((7)), and the continuous relaxation + DC penalty of §-E (box

((P2-C9) + (P2-DCP)) to (P2), we obtain the final problem (P3):

$$\begin{aligned}
& \min_{\{\mathbf{W}_k\}, \{\mathbf{S}_p\}, \{\mu_k\}, \{\mathbf{M}_p\}, \{b_{mp}\}} & \sum_{k=1}^K \text{tr}(\mathbf{W}_k) + \sum_{p=1}^P \text{tr}(\mathbf{S}_p) + \eta_b \sum_{m,p} (b_{mp} - b_{mp}^2) & \quad (\text{P3}) \\
& \text{s.t.} & \begin{bmatrix} \mathbf{A}_k + \mu_k \mathbf{I} & \mathbf{A}_k \hat{\mathbf{h}}_k \\ \hat{\mathbf{h}}_k^H \mathbf{A}_k & \hat{\mathbf{h}}_k^H \mathbf{A}_k \hat{\mathbf{h}}_k - \sigma_c^2 - \mu_k \epsilon_h^2 \|\hat{\mathbf{h}}_k\|^2 \end{bmatrix} \succeq \mathbf{0}, \quad \forall k & \quad (\text{P3-C1}) \\
& & \text{tr}(\mathbf{g}_p \mathbf{g}_p^H \mathbf{S}_p) \geq \gamma_S^{\text{PoD}} \sigma_s^2, \quad \forall p & \quad (\text{P3-C2}) \\
& & \begin{bmatrix} \mathbf{M}_p & \mathbf{I} \\ \mathbf{I} & \mathbf{J}_p^{\text{data}} \end{bmatrix} \succeq \mathbf{0}, \quad \forall p & \quad (\text{P3-C3}) \\
& & \text{tr}(\mathbf{M}_p) \leq \Gamma_{\text{Track},p}, \quad \forall p & \quad (\text{P3-C3b}) \\
& & \text{tr}(\mathbf{E}_m \mathbf{S}_p) \leq P_{\max} b_{mp}, \quad \forall m, p & \quad (\text{P3-C4'a}) \\
& & \text{tr}(\mathbf{E}_m \mathbf{R}_X) \leq P_{\max}, \quad \forall m & \quad (\text{P3-C4'b}) \\
& & \sum_{m=1}^M b_{mp} = N_{\text{req}}, \quad \forall p \in \mathcal{P}^{\text{active}} & \quad (\text{P3-C5'}) \\
& & \mathbf{W}_k \succeq \mathbf{0}, \quad \forall k & \quad (\text{P3-C6}) \\
& & \mathbf{S}_p \succeq \mathbf{0}, \quad \forall p & \quad (\text{P3-C7}) \\
& & \mu_k \geq 0, \quad \forall k & \quad (\text{P3-C8}) \\
& & 0 \leq b_{mp} \leq 1, \quad \forall m, p. & \quad (\text{P3-C9})
\end{aligned}$$

where $\mathbf{A}_k = \frac{1}{\gamma_k} \mathbf{W}_k - \sum_{j \neq k} \mathbf{W}_j - \sum_p \mathbf{S}_p$, $\mathbf{J}_p^{\text{data}}$ is an affine function of $\mathbf{S}_p$, and $\mathbf{E}_m$ is the antenna-block-extraction diagonal matrix. The feasible region of (P3) is the intersection of the LMI set (P3-C1), the LMI set (P3-C3), the linear equalities/inequalities (P3-C2)/(P3-C3b)/(P3-C4'a)/(P3-C4'b)/(P3-C5'), the positive-semidefinite cones (P3-C6)/(P3-C7), the scalar half-space (P3-C8), and the box constraints (P3-C9); each of these sets is convex, so the feasible region is convex. The objective $\sum_k \text{tr}(\mathbf{W}_k) + \sum_p \text{tr}(\mathbf{S}_p) + \eta_b \sum_{m,p} (b_{mp} - b_{mp}^2)$ is linear in $(\{\mathbf{W}_k\}, \{\mathbf{S}_p\})$ and is a “linear $-\eta_b \cdot$ quadratic” DC form in $\{b_{mp}\}$, and the overall objective is DC (non-convex) in $(\{\mathbf{W}_k\}, \{\mathbf{S}_p\}, \{b_{mp}\})$, so (P3) is not a convex SDP and its non-convexity is solved by the dual DC penalty SCA framework of §3.

Theorem 1 (Structural Properties of (P3)). *The feasible region of (P3) is convex; the objective is a DC function of $(\{\mathbf{W}_k\}, \{b_{mp}\}, \{\mathbf{S}_p\})$; the constraint matrices are affine in $(\{\mathbf{W}_k\}, \{\mathbf{S}_p\})$ and linear in $\{b_{mp}\}$. Hence (P3) is a difference-of-convex program (DCP), and a stationary point can be obtained by the dual DC penalty SCA framework of §3.*

Proof. See Appendix A. □

3 Joint DC Penalty SCA Framework: Rank-One and Binary Recovery

The feasible region of (P3) is convex but its objective is DC; obtaining the global optimum directly is hard, but *any stationary point of (P3) satisfying the Karush–Kuhn–Tucker (KKT) conditions is an engineering-acceptable solution.* This section designs a dual DC penalty SCA framework: the concave term $-\eta_b b_{mp}^2$ in the (P3) objective and the concave term $-\eta_{\text{rank}} \lambda_{\max}(\mathbf{W}_k)$ (corresponding to the missing rank-1 property) are linearized

simultaneously within the same SCA loop, and each iteration solves a standard convex SDP. This section also presents Gaussian randomization / leading-eigenvector extraction as a baseline post-processing.

A. Unified Characterization of the Dual DC Penalties

For any positive-semidefinite matrix $\mathbf{W}_k \succeq \mathbf{0}$, its trace equals its largest eigenvalue if and only if it has rank at most one. In the nonzero case (which is implied for an active user by a positive SINR target), this is equivalent to rank one:

$$\text{tr}(\mathbf{W}_k) - \lambda_{\max}(\mathbf{W}_k) = 0 \iff \text{rank}(\mathbf{W}_k) \leq 1. \quad (9)$$

Since $\lambda_{\max}(\mathbf{W}_k) \leq \text{tr}(\mathbf{W}_k)$ always holds for PSD matrices, define the rank-deficiency measure

$$\rho(\mathbf{W}_k) \triangleq \text{tr}(\mathbf{W}_k) - \lambda_{\max}(\mathbf{W}_k) \geq 0. \quad (10)$$

For $b \in [0, 1]$, define the binarity-deficiency measure

$$g(b) \triangleq b - b^2 = b(1 - b) \geq 0, \quad g''(b) = -2 < 0 \text{ (strictly concave)}. \quad (11)$$

$\rho(\mathbf{W}_k)$ and $g(b_{mp})$ are mathematically isomorphic—both are “convex – convex” DC forms, vanish on the desired constraint set, and are positive when the constraint is violated; this isomorphism is the mathematical foundation that allows the two penalties to share a single SCA framework. Adding the rank-1 penalty $\eta_{\text{rank}} \sum_k \rho(\mathbf{W}_k)$ to the (P3) objective gives

$$\min \quad \underbrace{\sum_k \text{tr}(\mathbf{W}_k) + \sum_p \text{tr}(\mathbf{S}_p)}_{\text{original objective}} + \underbrace{\eta_{\text{rank}} \sum_k \rho(\mathbf{W}_k)}_{\text{rank-1 penalty (DC)}} + \underbrace{\eta_b \sum_{m,p} g(b_{mp})}_{\text{binarity penalty (DC)}} \quad \text{s.t. (P3-C1)} \sim \text{(P3-C9)}. \quad (12)$$

All constraints of (P3-DCP) are convex (inherited from the convex feasible region of (P3)), and the only non-convexity comes from the two DC penalty terms in the objective.

B. SCA Linearization: Simultaneous First-Order Taylor Expansion of Both DC Terms

At iteration t , let the current iterate be $(\{\mathbf{W}_k^{(t)}\}, \{b_{mp}^{(t)}\})$. Perform the eigendecomposition

$$\mathbf{W}_k^{(t)} = \sum_{i=1}^{r_k} \lambda_{k,i} \mathbf{u}_{k,i} \mathbf{u}_{k,i}^H, \quad \lambda_{k,1} \geq \lambda_{k,2} \geq \dots \geq \lambda_{k,r_k} > 0, \quad (13)$$

where the leading eigenvector is $\mathbf{u}_{\max,k}^{(t)} \triangleq \mathbf{u}_{k,1}$. By the convexity of $\lambda_{\max}(\mathbf{W}_k)$, its first-order Taylor expansion provides a global lower bound:

$$\lambda_{\max}(\mathbf{W}_k) \geq \lambda_{\max}(\mathbf{W}_k^{(t)}) + \text{tr}(\mathbf{u}_{\max,k}^{(t)} (\mathbf{u}_{\max,k}^{(t)})^H (\mathbf{W}_k - \mathbf{W}_k^{(t)})) = \text{tr}(\mathbf{u}_{\max,k}^{(t)} (\mathbf{u}_{\max,k}^{(t)})^H \mathbf{W}_k). \quad (14)$$

Substituting this into the rank-1 penalty of (P3-DCP) yields a global upper bound (the concave term is replaced by its upper bound, the subproblem minimizes, and the true objective is a lower bound of the subproblem optimum):

$$\rho(\mathbf{W}_k) = \text{tr}(\mathbf{W}_k) - \lambda_{\max}(\mathbf{W}_k) \leq \text{tr}(\mathbf{W}_k) - \text{tr}(\mathbf{u}_{\max,k}^{(t)} (\mathbf{u}_{\max,k}^{(t)})^H \mathbf{W}_k) \triangleq \hat{\rho}^{(t)}(\mathbf{W}_k). \quad (15)$$

The first-order Taylor expansion of the concave $g(b) = b - b^2$ at $b^{(t)}$ gives the global overestimator

$$g(b) \leq g(b^{(t)}) + g'(b^{(t)})(b - b^{(t)}) = b^{(t)} - (b^{(t)})^2 + (1 - 2b^{(t)})(b - b^{(t)}), \quad (16)$$

and after removing the constant term $b^{(t)} - (b^{(t)})^2 - (1 - 2b^{(t)})b^{(t)} = (b^{(t)})^2$ (which does not affect minimization), the linearized penalty is

$$g(b) \leq (1 - 2b^{(t)})b + \text{const} \triangleq \hat{g}^{(t)}(b). \quad (17)$$

Substituting (15) and (17) into (P3-DCP) gives the convex SDP subproblem at iteration t :

$$\begin{aligned} \min_{\{\mathbf{W}_k\}, \{\mathbf{S}_p\}, \{\mu_k\}, \{\mathbf{M}_p\}, \{b_{mp}\}} \quad & \sum_{k=1}^K \text{tr}(\mathbf{W}_k) + \sum_{p=1}^P \text{tr}(\mathbf{S}_p) + \eta_{\text{rank}} \sum_{k=1}^K \left[\text{tr}(\mathbf{W}_k) - \text{tr}(\mathbf{u}_{\max,k}^{(t)} (\mathbf{u}_{\max,k}^{(t)})^H \mathbf{W}_k) \right] \\ & + \eta_b \sum_{m,p} (1 - 2b_{mp}^{(t)}) b_{mp} \end{aligned}$$

s.t. (P3-C1) $\sim$ (P3-C9).

(18)

The objective of (18) is linear in $(\{\mathbf{W}_k\}, \{\mathbf{S}_p\}, \{b_{mp}\})$ (the η_{rank} term combines $\text{tr}(\mathbf{W}_k)$ with $-\text{tr}(\mathbf{u}_{\max,k}^{(t)} (\mathbf{u}_{\max,k}^{(t)})^H \mathbf{W}_k)$ to form $\text{tr}((\mathbf{I} - \mathbf{u}_{\max,k}^{(t)} (\mathbf{u}_{\max,k}^{(t)})^H) \mathbf{W}_k)$, and the η_b term is linear in b_{mp}), and the constraints (P3-C1)–(P3-C9) are all LMI / linear / positive-semidefinite / box, so (P3-SCA- t) is a standard convex SDP that can be efficiently solved by interior-point solvers such as MOSEK/SDPT3. Update the iterate $(\{\mathbf{W}_k^{(t+1)}\}, \{b_{mp}^{(t+1)}\}, \{\mathbf{S}_p^{(t+1)}\}, \{\mu_k^{(t+1)}\})$ and repeat.

Theorem 2 (Dual DC SCA Convergence). *For fixed $\eta_{\text{rank}}, \eta_b > 0$, exact solution of every convex subproblem, and a feasible initial point, the true fixed-penalty objective $\{F_\eta^{(t)}\}$ generated by Algorithm 1 is monotonically non-increasing and bounded below, hence convergent. This statement alone does not assert that a finite penalty makes either residual exactly zero. If an accumulation point has $\rho(\mathbf{W}_k^*) = 0$ for every nonzero $\mathbf{W}_k^*$ and $g(b_{mp}^*) = 0$ for every (m, p) , then the rank-one extraction and the cardinality-preserving assignment agree with that point and satisfy the original constraints (P1-C1)–(P1-C7').*

Proof. See Appendix B. □

C. Baseline: Gaussian Randomization / Eigen-Extraction

If one pursues the fastest single-SDP run, the master problem (P3) with $\eta_{\text{rank}} = \eta_b = 0$ can be solved directly and a rank-one candidate can be extracted from its leading eigenvectors. This is only a baseline: extraction and assignment projection can break the robust-SINR/PCRB guarantees, so every candidate must be checked against the original constraints and an infeasible candidate must be reoptimized or discarded.

After the convexification chain of §2 and the dual DC penalty SCA of §3, all physical constraints of (P3) (robust SINR, PCRB, sensing SINR, AP power, AP-target assignment, box) are represented by convex constraints. The remaining rank and binarity residuals are controlled in the same SCA loop; their numerical values, the cardinality-preserving projection, and the subsequent physical feasibility check must be reported rather than inferred from an unverified exact-penalty claim.

Algorithm 1 Dual DC Penalty SCA Iteration (Joint Rank-One and Binary Recovery)

Require: Fixed penalty factors $\eta_{\text{rank}}, \eta_b > 0$, tolerance $\varepsilon > 0$, and maximum number of iterations T_{max} .

Ensure: An approximately rank-one covariance, a cardinality-feasible binary assignment, and a physically verified recovered solution when recovery succeeds.

- 1: Solve the unpenalized SDR ($\eta_{\text{rank}} = \eta_b = 0$) to obtain a feasible SCA start ($\{\mathbf{W}_k^{(0)}\}, \{b_{mp}^{(0)}\}, \dots$). Report infeasible if it has no solution. A heuristic may provide a warm start, but may affect the stationary point reached.
 - 2: **for** $t = 0, 1, \dots, T_{\text{max}} - 1$ **do**
 - 3: **for** each $k \in \mathcal{K}$ **do**
 - 4: Eigendecompose $\mathbf{W}_k^{(t)}$ to obtain the leading eigenvector $\mathbf{u}_{\text{max},k}^{(t)}$.
 - 5: **end for**
 - 6: **for** each (m, p) **do**
 - 7: Compute the g' coefficient $c_{mp}^{(t)} = 1 - 2b_{mp}^{(t)}$.
 - 8: **end for**
 - 9: Solve the convex SDP (18) to obtain $(\{\mathbf{W}_k^{(t+1)}\}, \{b_{mp}^{(t+1)}\}, \{\mathbf{S}_p^{(t+1)}\}, \{\mu_k^{(t+1)}\}, \{\mathbf{M}_p^{(t+1)}\})$.
 - 10: Compute $\delta_{\text{rank}}^{(t)} = \sum_k \rho(\mathbf{W}_k^{(t+1)})$ and $\delta_b^{(t)} = \sum_{m,p} g(b_{mp}^{(t+1)})$.
 - 11: Evaluate the true fixed-penalty objective $F_\eta^{(t+1)}$ in (12).
 - 12: **if** $\delta_{\text{rank}}^{(t)} \leq \varepsilon$, $\delta_b^{(t)} \leq \varepsilon$, and $|F_\eta^{(t+1)} - F_\eta^{(t)}| \leq \varepsilon$ **then**
 - 13: **break**.
 - 14: **end if**
 - 15: **end for**
 - 16: For each target p , set exactly the N_{req} largest entries of $\{b_{mp}^{(t+1)}\}_{m=1}^M$ to one and the others to zero, obtaining b_{mp}^{fix} .
 - 17: Re-solve the continuous fixed-assignment recovery problem with $b = b^{\text{fix}}$, and verify every original physical constraint. If $\delta_{\text{rank}} \leq \varepsilon$ and the recovered $\mathbf{W}_k$ is nonzero, set $\mathbf{w}_k^* = \sqrt{\lambda_{\text{max}}(\mathbf{W}_k)} \mathbf{u}_{\text{max},k}$; otherwise declare recovery unsuccessful.
 - 18: **return** only a physically verified recovered solution.
-

Algorithm 2 Rank-One Beamforming Extraction (Baseline Post-Processing)

Require: Converged solution $\{\mathbf{W}_k^*, \{\mathbf{S}_p^*\}, b_{mp}^*\}$.

Ensure: Rank-one beamforming vectors $\{\mathbf{w}_k^*\}$.

- 1: **for** each $k \in \mathcal{K}$ **do**
 - 2: Compute the eigendecomposition $\mathbf{W}_k^* = \mathbf{U}_k \mathbf{\Lambda}_k \mathbf{U}_k^H$.
 - 3: Extract the leading eigenvector: $\tilde{\mathbf{w}}_k = \sqrt{\lambda_{\text{max}}(\mathbf{\Lambda}_k)} \cdot \mathbf{u}_k^{\text{max}}$.
 - 4: Scale to satisfy the per-AP power: $\mathbf{w}_k^* = \eta_k \tilde{\mathbf{w}}_k$, where η_k is chosen so that $\text{tr}(\mathbf{E}_m \mathbf{R}_X) \leq P_{\text{max}}$ with $\mathbf{R}_X = \sum_k \mathbf{w}_k^* (\mathbf{w}_k^*)^H + \sum_p \mathbf{S}_p^*$ for every m .
 - 5: **end for**
 - 6: **if** $\{\mathbf{w}_k^*, b_{mp}^*\}$ satisfies all of (P1-C1)–(P1-C7') **then**
 - 7: **return** $\{\mathbf{w}_k^*, b_{mp}^*\}$.
 - 8: **else**
 - 9: Solve a fixed-assignment recovery problem and recheck all original constraints; if it is infeasible, discard this candidate.
 - 10: **end if**
-

A Proof of Theorem 1

This appendix proves Theorem 1.

The feasible region of (P3) is the intersection of the LMI set (P3-C1), the linear equalities/inequalities (P3-C2)/(P3-C3b)/(P3-C4'a)/(P3-C4'b)/(P3-C5'), the LMI set (P3-C3), the positive-semidefinite cones (P3-C6)/(P3-C7), the scalar half-space (P3-C8), and the box constraints (P3-C9). The LMI set, the linear-inequality/equality set, the positive-semidefinite cone, the scalar half-space, and the box set are all convex; the intersection of finitely many convex sets is convex, so the feasible region of (P3) is convex.

The objective $F(\{\mathbf{W}_k\}, \{b_{mp}\}, \{\mathbf{S}_p\}) = \sum_k \text{tr}(\mathbf{W}_k) + \sum_p \text{tr}(\mathbf{S}_p) + \eta_b \sum_{m,p} (b_{mp} - b_{mp}^2)$ is linear in $(\{\mathbf{W}_k\}, \{\mathbf{S}_p\})$ and is a “linear $-\eta_b \cdot$ quadratic” DC form in $\{b_{mp}\}$. The constraint matrices are affine in $(\{\mathbf{W}_k\}, \{\mathbf{S}_p\})$ and linear in $\{b_{mp}\}$, so (P3) is a DC program.

For complexity, the total number of decision variables is $n = K \cdot (MN_t)^2 + P \cdot (MN_t)^2 + K + P \cdot N_\theta^2 + MP$ (K matrices $\mathbf{W}_k \in \mathbb{H}_+^{MN_t}$, P matrices $\mathbf{S}_p \in \mathbb{H}_+^{MN_t}$, K S-Procedure relaxation variables μ_k , P auxiliary matrices $\mathbf{M}_p \in \mathbb{H}_+^{N_\theta \times N_\theta}$, and MP scalars b_{mp}); the number of inequality constraints is $m = K + P + P + P + MP + M + P_{\text{active}} + K + P + K + 2MP = 3K + 4P + M + 3MP + P_{\text{active}}$ ((P3-C1): K ; (P3-C2): P ; (P3-C3): P ; (P3-C3b): P ; (P3-C4'a): MP ; (P3-C4'b): M ; (P3-C5'): P_{active} ; (P3-C6): K ; (P3-C7): P ; (P3-C8): K ; (P3-C9): $2MP$); the largest LMI constraint size is $\ell = \max\{MN_t + 1, 2N_\theta\}$. Each SCA subproblem (P3-SCA- t) is a standard convex SDP, solvable in polynomial time by interior-point methods.

B Proof of Theorem 2

This appendix proves monotonicity of the true fixed-penalty objective; it does not claim exact rank-one or binary recovery for an arbitrary finite penalty.

Let $F_\eta(x)$ denote the objective in (12), where x collects all decision variables. At iteration t , let $Q_t(x)$ be its SCA majorizer obtained by replacing $\rho(\mathbf{W}_k)$ with (15) and $g(b_{mp})$ with the full affine upper bound in (16), including its constant term. The constants are omitted in (18) only because they do not change its minimizer.

For every feasible x , convexity of $\lambda_{\max}$ and concavity of g give $F_\eta(x) \leq Q_t(x)$. The two bounds are tight at the current iterate, so $F_\eta(x^{(t)}) = Q_t(x^{(t)})$. If $x^{(t+1)}$ minimizes the convex subproblem, then

$$F_\eta(x^{(t+1)}) \leq Q_t(x^{(t+1)}) \leq Q_t(x^{(t)}) = F_\eta(x^{(t)}).$$

Thus $\{F_\eta(x^{(t)})\}$ is monotonically non-increasing. It is bounded below by zero because the power and both deficiency measures are nonnegative; hence it converges.

At any accumulation point for which $\rho(\mathbf{W}_k^*) = 0$ and $\mathbf{W}_k^* \neq 0$, $\mathbf{W}_k^*$ is rank one and can be represented by its leading eigenvector. If also $g(b_{mp}^*) = 0$, then $b_{mp}^* \in \{0, 1\}$. In this special exact-recovery case, the convexification equivalence chain ensures the original constraints. For finite penalties, residuals must instead be reported and the cardinality-preserving projection followed by fixed-assignment recovery in Algorithm 1 must be physically verified.

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